Joonsuh Yoon

royyn04@gmail.com · appendix0.github.io · GitHub · LinkedIn

Education & Affiliations

Yonsei University — Undergraduate, Mechanical Engineering
GPA: **3.61 / 4.30** · Completed four semesters through Fall 2024
Currently on leave for military service · Planned return: March 2027

Republic of Korea Air Force — Enlisted service, Seoul Air Base, Mar 2025–present
Expected discharge: December 9, 2026

Research Interests

I am particularly interested in bringing AI agents into the real world through robotics. I aim to study how reinforcement learning can enable robots to acquire skills and operate autonomously in complex, open-ended environments.

Relevant Coursework & Research Preparation

UC Berkeley CS285: Deep Reinforcement Learning — Independent study
Completed the course and all homework assignments using the publicly available 2023 lectures and course materials.

Paper reading: Studied SkiMo, DreamSmooth, HumanoidBench, and other papers from SRL's recommended reading list.

Selected Projects

PincerCraft — LLM agent planning and evaluation in Minecraft May 2026

Independent project

- Implemented multistep planning, state feedback, action checks, and task-completion verification on Mindcraft, using existing large language models and action functions.
- Developed an evaluation system comparing agent configurations on 13 Minecraft tasks. With the same model, false completion—claiming completion without meeting task criteria—occurred in 2/52 attempts with the full harness and 23/51 with its experimental components removed.

Memex — Shared memory and associative trails for AI agents Aug 2026

Independent project

- Developed a shared knowledge library to address context fragmented across multiple agents, inspired by Vannevar Bush's *As We May Think*.
- Implemented record-selection and associative-trail workflows to connect findings and decisions; used and refined the system across personal projects over several months.

Additional Project Experience

TTNK / Jarada — Product design and development Jun–Aug 2026

Contributed to the design and development of a wearable device for posture feedback. The team received an award in the Republic of Korea Air Force startup competition.

Social Engagement Fund (SEF), Project participant — ATClink Oct 2026–present

Participate in a team project linking medication information across Korea, Vietnam, Nepal, and Uzbekistan through SEF, organized by Yonsei University's Institute for Global Engagement and Empowerment (IGEE).

Technical & Language Skills

Python / PyTorch: Implemented behavioral cloning, DAgger, policy gradients, DQN, SAC, and model-based RL within CS285 homework codebases.

Development & Infrastructure: JavaScript/Node.js, Linux, Git, LLM API integration, and cloud ARM server operation.

English: TOEIC 950/990 (August 2023). Comfortable reading and interpreting research papers in English without translation.

Honors & Awards

Republic of Korea Air Force Startup Competition — Award recipient (TTNK) Aug 2026
Incheon Inter-University Startup Competition — Excellence Award Sep 2024
Yonsei University Startup Support Foundation Competition — Award recipient Aug 2024